MCA-II

SCSS

Subject: Object Oriented Programming

Max marks:40

Time: Two hours

1(a) What is the need of Object-oriented Programming? How Java language full fill this objective? (5)

(b) What are different types of inheritance supported by Java?. Write brief code for each **(5)**

2. Write a static function lowestCommon that takes two long arguments and returns the position of the first set bit in common, where position 0 is the LSB. If there is no common bit, the function should return -1. For example lowestCommon(14,25) would be 3. Your solution should use at least one break statement. **(10)**

3(a). Write a function that accepts a string 'str' as its argument. The function needs to return the transformed string by replacing all occurrences of the character 'a' with the character 'b' and vice versa. **(5)**

(b) Write a program in java to show the function of comma, right shift and short-circuit logical operator. **(5)**

4. Write the purpose of "this" and super keyword by writing code. Write the purpose of "finalize" method by writing code. **(5)**

5. The function accepts an integer array 'arr' of size 'n' as its argument. The function needs to return the index of an

equilibrium point in the array, where the sum of elements on the left of the index is equal to the sum of elements on the right of the index. If no equilibrium point exists, the function should return -1. (10)

6. Write a function that accepts an integer array 'arr' of size 'n' and an integer 'd' as its argument. The function needs to rotate the array 'arr' by 'd' positions to the right. The rotation should be done in place, without using any additional memory. (10)

3 2 1 5 4

d=3

MCA-II semester

CS-410(Object Oriented Programming)

Time: 3 hours

Max. Marks: 60

Note: **Write answer of any six questions. All questions carry equal marks.**

1. Imagine you're tasked with designing a highly concurrent application in Java that processes a large dataset. The application needs to meet the following requirements:

Utilize multiple threads efficiently to process different parts of the dataset concurrently.

i. Implement a mechanism to pause and resume threads dynamically based on system load or user input.

ii. Ensure thread safety and prevent data corruption without relying solely on synchronized blocks or locks.

Design a solution that incorporates the following keywords and concepts from Java's multithreading:

Volatile, wait(), notify(), Thread.sleep() Atomic classes (e.g., AtomicInteger, AtomicBoolean) CountDownLatch or CyclicBarrier

Your solution should address the requirements effectively while discussing the trade-offs and advantages of each keyword or concept used. Provide a detailed explanation of how your solution ensures thread safety, scalability, and responsiveness in handling the dataset processing.

2. Consider a scenario where you have a complex inheritance hierarchy representing different types of vehicles. Each vehicle type has specific attributes and behaviors, and there are multiple levels of inheritance involved. Your task is to design and implement a Java program that models this hierarchy effectively. Discuss how you would structure the classes, handle inheritance, and ensure proper encapsulation and abstraction. Provide a code example demonstrating your approach, including the definition of at least three levels of inheritance and the implementation of overridden methods with polymorphic behavior. Write Implementation in java programming.

vehicle()
~~int wheels~~
String name;
Model no;
runs( )
{
}

3. In today's rapidly evolving technology landscape, where new programming languages and frameworks emerge frequently, why does Java programming continue to be relevant and widely used in various industries and applications? Discuss the unique features, advantages, and strengths of Java that contribute to its enduring popularity, and compare its capabilities with those of other programming languages. Additionally, explore the challenges and limitations of Java in addressing modern software development needs and how these challenges can be mitigated or overcome. Provide real-world examples or case studies to support your arguments and demonstrate Java's relevance in contemporary software development practices.

4. You're tasked with developing a robust file processing module in Java that reads data from multiple files, processes it, and writes the results to an output file. However, various exceptions may occur during file I/O operations, such as FileNotFoundException, IOException, and others. Discuss how you would design an exception handling mechanism to gracefully handle these exceptions, provide meaningful error messages to users, and ensure data integrity. Additionally, consider scenarios where the input files may be large or numerous, and outline strategies to optimize performance and memory usage while maintaining robust exception handling. Write a Java program to implement it.

5. You're tasked with developing a program that simulates a complex traffic intersection with multiple lanes and traffic lights. Each lane has its own set of traffic lights controlling the flow of vehicles. Design and implement a Java program that models this intersection, incorporating loop structures and control statements to manage the traffic flow effectively. Consider various scenarios, such as peak traffic hours, emergency vehicle prioritization, and pedestrian crossings. Discuss how you would use loops to simulate the passage of time and control statements to enforce traffic rules and signal changes. Additionally, explore strategies for optimizing the efficiency of the intersection, handling edge cases, and ensuring safety and compliance with traffic regulations. Provide a detailed explanation of your approach, including the design of data structures, algorithms, and the logic for managing traffic flow in a complex urban environment. Write a Java program to implement it.

6. When you extend a class, the constructor for your new class will reference the constructor of the parent class, and this latter constructor may have any of the four possible access regimes. Comment on the consequences of each of the four possibilities by writing suitable code in java.

1. Write TRUE and FALSE for the following statements:

a. All methods in an abstract superclass must be declared abstract. +
b. A class declared final cannot be subclassed.
c. Every method of a final class is implicitly final.
d. A redefinition of a superclass method in a subclass need not have the same signature as the superclass method. Such a redefinition is not method overriding but is simply an example of method overloading.
e. A constructor is a special method with the same name as the class that is used to initialise the members of a class object. Constructors are called when objects of their classes are instantiated.
f. A method declared static cannot access non-static class members. A static method has no this reference because static class variables and static methods exist independent of any objects of a class.
g. An array subscript may be an integer or an integer expression. If a program uses an expression as a subscript, then the expression is evaluated to determine the particular element of the array.
h. To pass one row of a double-subscripted array to a method that receives a single-subscripted array, simply pass the name of the array followed by the row subscript.
i. Overloaded methods can have different return values, and must have different parameter lists. Two methods differing only by return type will result in a compilation error.
j. Any block of Java code may contain variable declarations.